JUAN ESTEBAN VILLADA SIERRA

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PROFESSIONAL SUMMARY

Electronic Engineering student (UNAL) profiling as an **Edge AI & Audio Fullstack Engineer**. Solid experience in hardware-software integration, with a portfolio spanning hardware descriptions in FPGAs (Verilog) and RTOS/Linux architectures, to the development of immersive, high-performance web interfaces (React 19, WebGL/Three.js) and the orchestration of LLM agents (Serverless and Offline). Specialist in designing robust, low-latency architectures to solve complex problems at the edge.

EDUCATION

Universidad Nacional de Colombia (UNAL) — Manizales, Colombia

B.Sc. Electronic Engineering — Expected Graduation: Dec 2026

- **Relevant Coursework:** Real-Time Systems, Dynamic Systems and Control, Digital Signal Processing (DSP), Artificial Intelligence, Embedded Systems.
- **Activities:** Active member of the AI and Audio Synthesis Research Groups (UNAL).

WORK EXPERIENCE

Software & AI Developer — Jan 2025 – Present

AI Research Group, Universidad Nacional de Colombia — Manizales, Colombia

- Led the development of a vehicle license plate classification system using computer vision (Python/ML), achieving **95% inference accuracy**.
- Integrated SQL databases for the efficient management and monitoring of detected records.
- The project was successfully **scaled and implemented by the Colombian National Police** for urban surveillance systems.

TECHNICAL PROJECTS

Hardware DSP Bass Synthesizer (FPGA) — 2026

Verilog, Gowin Tang Nano 20K, Faust, I2S

- Designed and implemented a digital bass synthesizer using physical modeling (Karplus-Strong) completely in hardware.
- Prototyped the algorithm in Faust DSP and manually translated the mathematics to Q15 fixed-point arithmetic in pure Verilog, achieving **2.1ms latency on a 27MHz clock** with no OS overhead and direct 24-bit I²S DAC output.
- Implemented advanced **anti-overflow and limit-cycle mitigation techniques** for low-pass filters and soft-clipping saturators.

Linux Embedded / Lock-free C++ Audio Engine — 2025

C++, ALSA, FluidSynth, ESP32, FreeRTOS, Raspberry Pi Zero 2W

- Engineered a hybrid asymmetric hardware synthesizer: an ESP32 (FreeRTOS) scanning a custom key matrix and a Raspberry Pi running a custom C++ audio engine.
- Developed **lock-free queues in C++** to achieve deterministic, real-time communication between the MIDI parsing thread and the ALSA audio callback, minimizing jitter.
- Optimized FluidSynth compilation on embedded Linux for headless polyphonic synthesis.

High-Performance WebGL & Serverless AI (Antioquia Zana 3D) — 2026

React 19, Three.js, FastAPI, Vercel

- Developed interactive 3D SPAs integrating advanced graphics with React Three Fiber, GLSL shaders, and glassmorphism.
- Optimized the GPU's garbage collection and lazy loading to maintain a **rock-solid 60 FPS** without memory leaks.
- Deployed a serverless Python backend (FastAPI) on Vercel to orchestrate AI agents (Llama 3.3 70B via Groq API), implementing warm starts for ultra-low latencies (<800ms).

Offline Edge AI Multi-Agent System (RestauranteIA) — 2025

Python, LangChain, Ollama, Llama 3.1, NVIDIA Jetson

- Developed a 100% offline multi-agent system using local models via Ollama.
- Implemented **native Tool Calling with strict Pydantic typing** and SQLite to eliminate hallucinations within the AgentExecutor loop.
- Deployed facial recognition computer vision models directly on NVIDIA Jetson platforms, enabling immediate hardware inference without cloud dependency.

TECHNICAL SKILLS

Languages:	C/C++, Embedded C, Verilog, Python, TypeScript/JavaScript, Assembly, SQL.

Frontend & WebGL:	React 19, Three.js (React Three Fiber / Drei), Tailwind CSS v4, Zustand, Vite.
Backend & AI:	FastAPI, Vercel Serverless, Pydantic, Groq API, LangChain, Ollama (Llama 3), PyTorch.
Embedded Systems:	RTOS (FreeRTOS), Embedded Linux (POSIX, Systemd), DSP (Fixed-Point), OTA, I2S.
Hardware:	FPGAs (Gowin), ESP32, Raspberry Pi, NVIDIA Jetson, PCB Design (Altium).
Software & Tools:	Git/GitHub, Docker, Linux (Bash), JUCE Framework, Ableton Live.

CERTIFICATIONS & LANGUAGES

Certification:	Deep Learning Specialization - Coursera / DeepLearning.AI
Languages:	Spanish (Native), English (Advanced - B2 Level) - Fluent technical and conversational competence.